NetworkAI-Gym: A Reliable Networking-Based Platform For Reinforcement Learning Researchers

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Abstract

Low Earth Orbit (LEO) satellite networks are critical to Next-Generation Network (NGN) technologies, providing extensive wireless access and supporting diverse applications. However, high launch costs and short lifespans pose challenges, with satellites lasting approximately five years and launches costing. Efficient energy management, particularly RL-based power control, can extend satellite lifespan and reduce costs. Due to computational constraints of satellite, a reliable simulation platform is required for the RL model pre-train. To address the lack of simulation platforms for satellite power control, we introduce NetworkAI-Gym, a simulation environment designed for developing and testing RL-based algorithms for satellite energy management, enabling efficient and sustainable network operations.

1 Introduction

The LEO satellite network, is a vital component of Next-Generation Network (NGN) technologies [1]. The LEO satellites are located at altitudes between 200 km and 1200 km, and move at about 7.56 km/s, offering radio coverage of up to 100-200 km in radius [2]. An LEO satellite can provide massive wireless access to ground users and support different user applications. However, high launch costs and short satellite lifespan of the satellite pose a significant challenge for operators: (1) Each Falcon 9 launch carries Starlink satellites valued at approximately 17 to 18 million dollars [3]. (2) A Starlink satellite has a lifespan of approximately 5 years [4].

To extend the lifetime of the satellite and indirectly reduce monetary costs, one method is to reduce battery charging times by optimizing energy management systems (e.g., power control) onboard the satellites. Efficient power control can minimize the wear and tear on batteries, which are critical components of satellite operations. For example, reducing non-essential payload activity during periods of low power availability can further conserve battery life [5].

Nowadays, the power control issue plays an crucial role in energy management. Many researches discussed the intelligent RL-based power control. Nevertheless, a satellite has limited computational resources [6], which poses a significant challenge in implementing resource-intensive RL-based power control methods, and train within the satellite. Due to constraints such as power consumption, memory, and processing capabilities, RL algorithm acts as the “Table Look Up” will be the best solution for satellite power control.

It requires a reliable simulation platform to pre-train and validate the RL-based power control policies before deploying them to the satellite. By simulating real-world scenarios, including varying orbital dynamics, power demands, and environmental conditions, the platform can generate optimized policies that are efficient and lightweight enough for onboard execution.

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We notice that there are no platforms like OpenAI Gym [7] for addressing network issues (specifically for satellite power control). In this work, we introduce NetworkAI-Gym, a reliable simulation platform designed to support the development and testing of RL algorithms for handling complex networking challenges.

2 Related Works

This section has the literature survey for the previous works to satellite networking issues. We investigate the simulation of each work, and considering their simulation with three aspects:

Reliability: The simulation platform should accurately reflect real-world scenarios, including orbital dynamics, population distribution, and environmental conditions, to ensure the results are representative and applicable to actual satellite operations.

User-Friendly: The platform should offer an intuitive interface and straightforward configuration options, enabling researchers to easily design, test, and modify simulation scenarios without extensive setup or specialized knowledge.

Modular: The simulation platform should support modularity, allowing components set up basic network function, e.g., resource allocation, channel models, etc., to be independently developed, integrated, and tested, facilitating flexibility and scalability for various use cases.

Ran *et al.* [8] explore beam hopping and power allocation in satellite networks, employing reinforcement learning (RL)-based methods to address these challenges. Huang *et al.* [9] investigate Deep RL-based approaches for power allocation within satellite networks, while in another study, Huang *et al.* [10] focus on resource allocation using a similar Deep RL-based methodology. Tsuchida *et al.* [11] utilize Q-Learning to enable intelligent power control for satellites. Lee *et al.* [12] propose a Deep RL-based approach specifically aimed at optimizing power allocation.

However, the lack of standardization in simulation environments results in significant variation across studies, making fair algorithm comparisons difficult. Moreover, many authors fail to incorporate realistic scenario settings, such as actual satellite mobility patterns or real-world population distributions, into their simulations. This oversight raises concerns about the practicality and applicability of the proposed solutions in real-world deployments.

An increasing number of researchers are utilizing RL algorithms as tools to address networking challenges. To the best of our knowledge, there is no platforms like OpenAI Gym for addressing network issues. We construct a simulation platform named NetworkAI-Gym, and our contribution can be summarized as follows.

Contributions:

1. NetworkAI-Gym is an modular environment. Develop can easily install network functions (or models) based on different industry standard (e.g., channel model, communication model, etc). We install the basic network function like channel model and resource allocation mechanism following the 3GPP Standard [13] [2].
2. We import the real-word data: map, satellite information (e.g., constellation, trajectory), population distribution to enhance the reliability.
3. We create an RL tool box and implement the basic RL algorithms like Deep Q-Network. Developer can easily import the RL algorithm for different networking issue.
4. We give a guideline to the operation of NetworkAI-Gym to enhance the user-friendly.

3 NetworkAI-Gym Architecture

This section introduces the details of the NetworkAI-Gym platform. There are three cores in the NetworkAI-Gym platform:

[2]3GPP standards are the industry specifications and provide a comprehensive framework for designing, implementing, and evaluating telecommunication systems. These standards ensure interoperability between devices and networks, enabling the simulation to reflect real-world operational conditions accurately

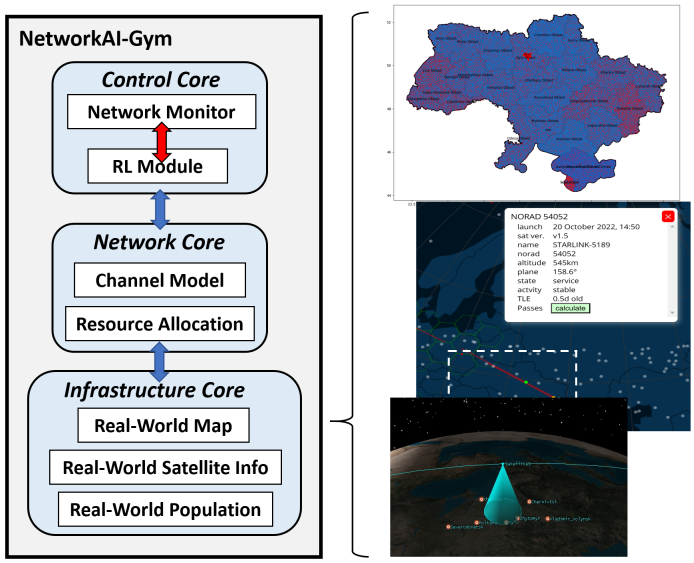

Figure 1: Overview of NetworkAI-Gym

- **Control Core**: The control core consists of two main components: the network monitor and the RL module. The network monitor collects performance metrics (e.g., satellite power levels, channel capacity, etc.) across the entire satellite network. These collected metrics enable developers to define the network state and design the reward function for the RL module.

  We establish seamless interaction between the network monitor and the RL module. Developers can easily import RL algorithms and interact with the satellite network, enabling efficient integration and experimentation.

- **Network Core**: The network core includes all the necessary physical models, such as the channel model, path loss model, and communication model, which are used to simulate data transmission. These models are based on the 3GPP standard to ensure accuracy and alignment with real-world telecommunications scenarios. The network core guarantees the real-world communication behaviors of satellite networks.

- **Infrastructure Core**: The network core is designed to set up and simulate satellite networks with enhanced realism. To achieve this, we incorporate real-world data, such as geographical maps (e.g., the Ukraine map), population distribution (e.g., Ukraine's population), and satellite constellation information (e.g., Starlink satellite constellation). Developers can utilize these data inputs to customize and create diverse network scenarios, enabling simulations that closely reflect real-world conditions.

3.1 Developer Guideline

To use NetworkAI-Gym, developers can follow the following steps:

1. *select_map() & select_satellite()*: This step allows users to choose a real-world map (e.g., country-specific maps such as the Ukraine map) and satellite constellation (e.g., Starlink) to define the simulation environment.
2. *load_rl_module*: Users can import a pre-existing RL algorithm or define their own. The RL module serves as the decision-making engine for network control tasks, such as satellite power control, resource allocation, or routing optimization.
3. *model_train()*: Train the RL model using the selected network scenario and predefined performance metrics.
4. *model_test()*: Evaluate the trained RL model in the simulation environment to assess its performance under various conditions.

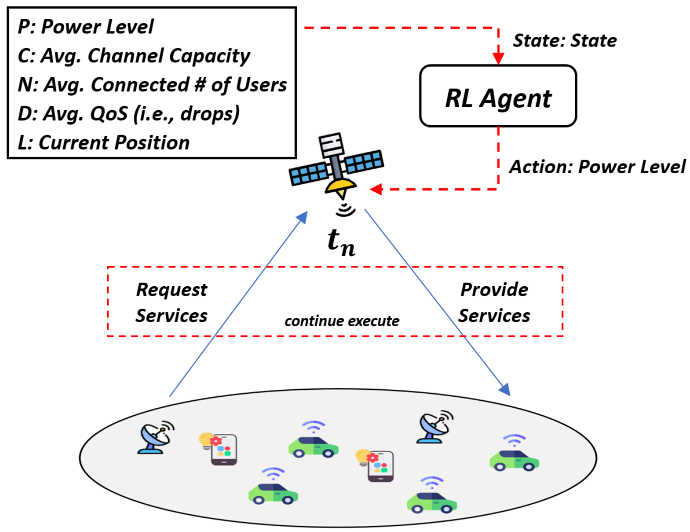

Figure 2: Satellite Intelligent Power Control Mechanism

4 Case Study: Intelligent Power Control

In this section, we provide a case study of applying RL to achieve satellite intelligent power control.

Figure 2 illustrates the power control mechanism. At each time step t, the satellite collects global network metrics, including:

(M1) the power level $P_t \in \mathbb{R}$.

(M2) the average channel capcacity $C_t \in (0, 1)$.

(M3) the average number of connected users $N_t \in \mathbb{R}$.

(M4) the average QoS $D_t \in \mathbb{R}$.

(M5) the current position of satellite $L_t = (x_t, y_t, z_t)$ in spherical coordinates.

We combine the above metrics as a network state of time step T, denoted as S_t:

$$S_t = \Big(P_t, C_t, N_t, D_t, L_t\Big) \tag{1}$$

The state space $\mathcal{S}$, which is an infinite state space.

Based on the network state S_t, the RL agent generates a power control decision to adjust the satellite's power level. The control decision at time step t is represented as $A_t \in \{1, 2, 3\}$, where each value corresponds to the number of power increase steps.

For state $S_t = s$, the probability of the RL agent taking a specific control decision $A_t = a$ is.

$$\pi(a|s) = \Pr[A_t = a|S_t = s] \tag{2}$$

where $\pi(a \mid s)$ denotes the policy that maps the state s to the probability of selecting action a.

After the RL agent taking the control decision A_t, it receives a reward, as feedback, denoted as R_t. The reward R_t is calculated by the following reward function.

$$R_t = (-1) \cdot \Big[\alpha_1 \cdot P_t + \alpha_2 \cdot D_t\Big] \tag{3}$$

where $\alpha_1 \in (0, 1)$, $\alpha_2 \in (0, 1)$ are the weight, and $\alpha_1 + \alpha_2 = 1$.

4.1 Problem Formulation

The control decision directly influences the observations, specifically M2 (average channel capacity), M3 (average number of connected users), and M4 (average QoS). For example, when the satellite operates at a high power level, fewer resources are required to maintain high QoS, making it suitable for scenarios with a high number of service requests. Conversely, when the satellite operates at a low power level, more resources are needed to achieve comparable QoS levels.

Considering the long-term benefit, at the end of the time step t, we define G_t as the accumulated discounted reward, and

$$G_t = R_t + \gamma \cdot R_{t+1} + \gamma^2 R_{t+2} + ... = \sum_{m=0}^{\infty} \gamma^m R_{t+m} \tag{4}$$

where $\gamma \in [0, 1]$ is the discount factor.

While we apply the policy π, we let $Q^\pi(s, a)$ be the conditional expected accumulated reward starting from $S_t = s$, taking the action $A_t = a$, and then $Q^\pi(s, a)$ is expressed as

$$Q^\pi(s, a) = \mathrm{E}\Big[G_t | S_t = s, A_t = a\Big] \tag{5}$$

Our goal is to find the optimal policy π^* such that we have the optimal $Q^*(s, a)$, that is,

$$Q^*(s, a) \doteq \max_\pi Q^\pi(s, a) = Q^{\pi^*}(s, a) \tag{6}$$

Based on the above discussion, we employ the Deep Q-Network (DQN) algorithm [14] in the proposed DRL model. DQN has been widely adopted to address reinforcement learning problems where the state space is continuous or infinite, and the action space is finite [15].

DQN utilizes a function approximator, specifically a neural network (referred to as the Q-Network), denoted by $Q(s, a|\theta)$, to approximate the optimal action-value function $Q^*(s, a)$. Here, θ represents the set of neural network parameters, including the weights and biases.

$$Q(s, a|\theta) \approx Q^*(s, a) \tag{7}$$

4.2 Performance Evaluation

In this section, we have the two main discussion:

(1) Can DQN solve the satellite power control?

We trained the DQN model and utilized the trained model for evaluation. We train the DQN with setting: the architecture of the value network is $[128, 128, 128, 128]$. We use a replay buffer of size 20480 and a batch size of 1024. The learning rate is 0.0005, and the discount factor is 0.99. The number of training episode is 3000.

Effect of the State Normalization

In many previous works, state normalization has often been applied with little explanation. Here, we offer a deeper discussion by comparing two DQN models: one with state normalization and one without. For the normalized model, each state component is scaled to the range of $[0, 1]$.

Fig. 3 shows the learning curve of the DQN model. The curve in blue represents the DQN model with state normalization and the curve in orange represents the DQN model without state normalization.

Initially, both models exhibit a sharp increase in accumulated reward, indicating rapid learning and optimization of actions during the early stages of training. As training progresses, the curves plateau, showing convergence toward a stable policy.

The shaded regions around the curves represent the standard deviation, highlighting the variability in rewards across episodes. However, the two curves overlap significantly, indicating comparable

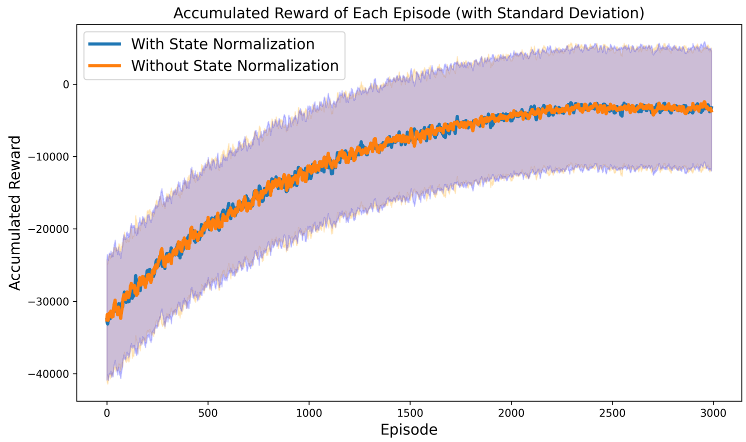

Figure 3: The Learning curve of the DQN (State Normalization v.s without State Normalization)

overall performance. We observed that the standard deviations of the reward curves for both models are approximately equivalent, measuring $8,294$ and $8,303$.

In summary, for the power control problem, state normalization does not significantly impact the learning process in rewards, as evidenced by the nearly identical standard deviations.

Evaluation Results

The trained DQN (with state normalization) model will be evaluated for 100 episode. We compared the DQN against two baseline methods: Round-Robin, which distributes the power level selection evenly, and Random, which selects actions arbitrarily without a specific strategy. Table. 1 shows the results of the mean reward of each episode, standard deviation, and variance.

We found that the DQN has the highest obtained reward with -0.75×10^2, with the lowest standard deviation 1.46×10^{-1}, and variance 2.1×10^{-2} .

Table 1: Performance Comparison

	Round-Robin	Random	DQN
Mean	-3.24×10^5	-3.26×10^5	-0.75×10^2
Std	0	1.81×10^3	1.46×10^{-1}
Variance	0	3.3×10^6	2.1×10^{-2}

In summary, the DQN model proves to be highly effective for managing intelligent power control in satellite networks, balancing energy efficiency with performance metrics such as QoS and resource allocation.

(2) What component significantly impact real-world implementation?

In many previous works, the satellite network often assume the population distribution is uniform.

The direct deployment of a trained DQN model on satellites may depend on a main factor: adaptability to real-world scenarios. While the DQN model performs well in simulated environments, real-world conditions such as unexpected interference, dynamic orbital changes, and population differ significantly.

In many previous works, the satellite network often assume the population distribution is uniform.

We investigate the impact of *population* distribution on the learning process and model performance, analyzing whether population scenarios influence the outcomes. We want to provide developers with deeper insights into the components of NetworkAI-Gym that are critical for training RL models tailored to specific issues.

Effect of the Population

Fig. 4 illustrates the learning curves for two scenarios: training with a uniform population distribution (red curve) and training with a real-world population distribution (blue curve).

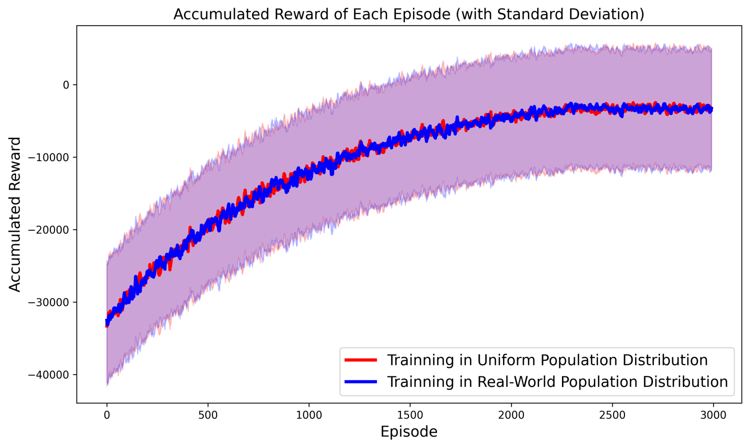

Figure 4: The Learning curve of the DQN (Uniform v.s Real-World Population Distribution)

Both scenarios exhibit similar trends, with rapid improvement in accumulated rewards during the early training stages, followed by stabilization as the models converge to a near-optimal policy.

This indicates that for the power control task examined here, the choice between a uniform or real-world population distribution does not significantly impact the RL model's performance.

These findings suggest that while realistic population data can enhance simulation fidelity, it may not always be a critical factor for certain tasks like power control. In other words, previous works assume the population is uniformly distributed is available is power control issue.

Due to space limitations, we focused exclusively on the effect of population distribution in this study. We will explore the impact of other components of NetworkAI-Gym, such as the channel model and communication mechanisms, to highlight the significance of these elements in developing effective RL-based solutions.

5 Future Works

While the current version of NetworkAI-Gym demonstrates its effectiveness in simulating satellite networks and enabling RL-based power control solutions, there are several avenues for future enhancement. These improvements aim to broaden the platform's applicability and effectiveness in real-world settings. Below, we outline some promising directions:

- **RL Algorithm Extension:** Future releases can include more diverse RL algorithms (e.g., policy gradients, actor-critic methods) to handle varying levels of complexity, state or action spaces, and reward structures, providing developers with a broader toolkit for tackling evolving network challenges.
- **Multi-Satellite and Multi-Agent Support:** Extending the platform to handle multiple satellites with multi-agent RL can better mirror real-world constellations and coordinated resource allocation.
- **Efficient Computation:** Techniques like distributed training and model compression can scale simulations and expedite experimentation.
- **Broader Use Cases:** NetworkAI-Gym's modular architecture allows exploration of beam hopping, routing, handover, and other network challenges beyond power control.

Future enhancements will strengthen NetworkAI-Gym as a versatile tool for researchers, aiding in the advancement of satellite network technologies.

6 Conclusion

We introduced NetworkAI-Gym, a simulation platform tailored to the development and testing of RL-based satellite network solutions. By incorporating real-world satellite constellations, population distributions, and 3GPP-based models, NetworkAI-Gym offers a realistic, modular, and user-friendly environment. Through a case study on satellite power control using DQN, we showed that DQN outperforms heuristic baselines, underscoring the potential of RL-based approaches to optimize power usage while meeting QoS demands.

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